

Enunciado:

Utilizando la teoría de diagonalización, hallar el valor de A^{100} siendo $A = \begin{pmatrix} 0 & 2 \\ -3 & 5 \end{pmatrix}$.

PDF

1. Obtener los autovalores de A

Calculamos $\det(A - \lambda I) = 0$:

$$\begin{vmatrix} -\lambda & 2 \\ -3 & 5 - \lambda \end{vmatrix} = (-\lambda)(5 - \lambda) - (-6) = \lambda^2 - 5\lambda + 6 = 0$$

Factorizando:

$$(\lambda - 2)(\lambda - 3) = 0 \implies \lambda_1 = 2, \quad \lambda_2 = 3$$

2. Obtener los autovectores asociados

Para $\lambda_1 = 2$:

$$(A - 2I)v = 0 \implies \begin{pmatrix} -2 & 2 \\ -3 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies -2x + 2y = 0 \implies x = y$$

Un autovector es $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

Para $\lambda_2 = 3$:

$$(A - 3I)v = 0 \Rightarrow \begin{pmatrix} -3 & 2 \\ -3 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow -3x + 2y = 0 \Rightarrow y = \frac{3}{2}x$$

Tomando $x = 2$, un autovector es $v_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

3. Matrices P , D e inversa P^{-1}

Matriz diagonal D :

$$D = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

Matriz de paso P :

$$P = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}$$

Inversa P^{-1} :

$$\det(P) = (1)(3) - (2)(1) = 1$$

$$P^{-1} = \frac{1}{1} \begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix}$$

4. Calcular $A^{100} = P \cdot D^{100} \cdot P^{-1}$

$$D^{100} = \begin{pmatrix} 2^{100} & 0 \\ 0 & 3^{100} \end{pmatrix}$$

Multiplicamos paso a paso:

$$P \cdot D^{100}:$$

$$\begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 2^{100} & 0 \\ 0 & 3^{100} \end{pmatrix} = \begin{pmatrix} 2^{100} & 2 \cdot 3^{100} \\ 2^{100} & 3 \cdot 3^{100} \end{pmatrix}$$

$$(P \cdot D^{100}) \cdot P^{-1}:$$

$$A^{100} = \begin{pmatrix} 2^{100} & 2 \cdot 3^{100} \\ 2^{100} & 3 \cdot 3^{100} \end{pmatrix} \begin{pmatrix} 3 & -2 \\ -1 & 1 \end{pmatrix}$$

$$A^{100} = \begin{pmatrix} 3 \cdot 2^{100} - 2 \cdot 3^{100} & -2 \cdot 2^{100} + 2 \cdot 3^{100} \\ 3 \cdot 2^{100} - 3 \cdot 3^{100} & -2 \cdot 2^{100} + 3 \cdot 3^{100} \end{pmatrix}$$